Worksheet 4 solution - Curves and integration.

1. Consider the curve C in $\mathbb{R}^3$ given by

$$\mathbf{r}(t) = \left(e^t \cos t\right)\mathbf{i} + 2\mathbf{j} + \left(e^t \sin t\right)\mathbf{k}$$

(a) Draw a sketch of C.

Solution. Since the y coordinate of $\mathbf{r}(t)$ is fixed at 2, this is basically plane curve. The x and z coordinates describe an exponential spiral where the radius at angle t is e^t. Thus, a sketch of C is the following graph.

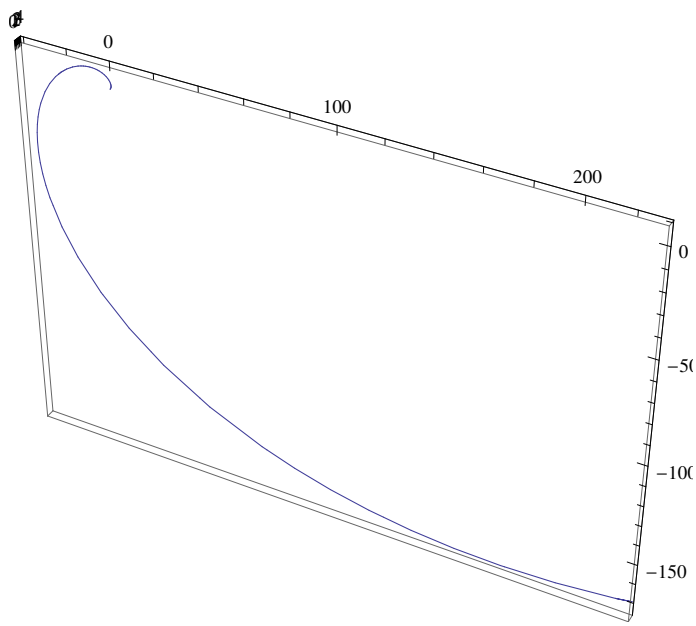

Figure 1: Sketch of C.

(b) Calculate the arc length function $s(t)$, which gives the length of the segment of C between $\mathbf{r}(0)$ and $\mathbf{r}(t)$ as a function of the time t for all $t \geq 0$. Check your answer with the instructor.

Solution. Since

$$x'(t) = e^t \cos t - e^t \sin t, \qquad y'(t) = 0, \qquad z'(t) = e^t \sin t + e^t \cos t,$$

we have

$$|\mathbf{r}'(t)| = \sqrt{(e^t \cos t - e^t \sin t)^2 + (e^t \sin t + e^t \cos t)^2} = \sqrt{2}e^t.$$

Hence the arc length is

$$s(t) = \int_0^t |\mathbf{r}'(u)|\,du = \int_0^t \sqrt{2}e^u\,du = \sqrt{2}e^t - \sqrt{2}.$$

(c) Now invert this function to find the inverse function $t(s)$. This gives time as a function of arclength, that is, tells how long you must travel to go a certain distance.

Solution. Solve $s = \sqrt{2}e^t - \sqrt{2}$, which gives $e^t = \frac{s+\sqrt{2}}{\sqrt{2}}$, and so

$$t = t(s) = \ln\left(\frac{s+\sqrt{2}}{\sqrt{2}}\right).$$

2. Consider the curve C given by the parametrization $\mathbf{r}\colon \mathbb{R} \to \mathbb{R}^3$ where $\mathbf{r}(t) = (\sin t, \cos t, \sin^2 t)$.

(a) Show that C is in the intersection of the surfaces $z = x^2$ and $x^2 + y^2 = 1$.

Solution. Since $x = \sin t, y = \cos t, z = \sin^2 t$, it is very easy to check that $z = x^2$ and $x^2 + y^2 = 1$. So the curve C lies in both these two surfaces, hence is in the intersection of them.

(b) Use (a) to help you sketch the curve C.

Solution. The left graph is the intersection of the two surfaces, while the right one is the curve.

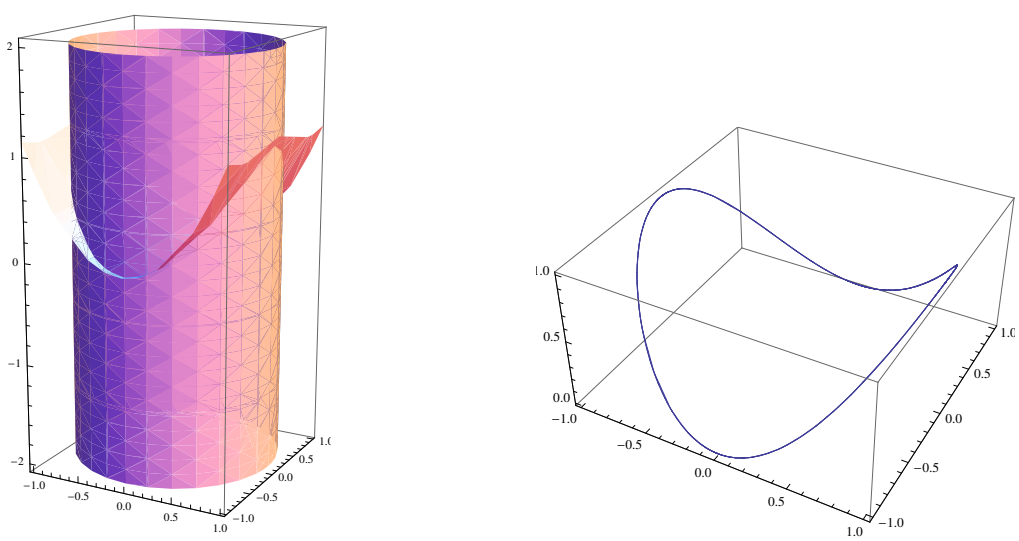

Figure 2: Two surfaces and the curve C.

3. (a) Sketch the top half of the sphere $x^2 + y^2 + z^2 = 5$. Check that $P = (1, 1, \sqrt{3})$ is on this sphere and add this point to your picture.

Solution. The top half of the sphere is shown in Figure 3 (the black dot is P). Since $1^2 + 1^2 + (\sqrt{3})^2 = 5$, we know P is on this sphere.

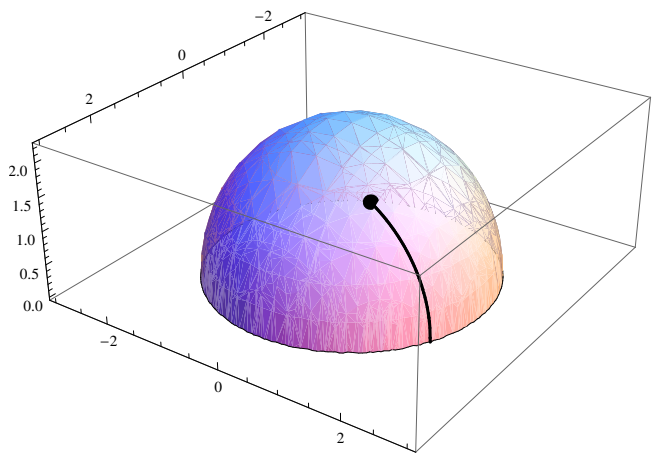

Figure 3: Half sphere and the path.

(b) Find a function $f(x, y)$ whose graph is the top-half of the sphere. Hint: solve for z.

Solution. Since $x^2 + y^2 + z^2 = 5$, we have $z^2 = 5 - x^2 - y^2$, and so $z = \pm\sqrt{5 - x^2 - y^2}$. As we only want the top half of the sphere, we can let $f(x, y) = \sqrt{5 - x^2 - y^2}$.

(c) Imagine an ant walking along the surface of the sphere. It walks *down* the sphere along the path C that passes through the point P and stays parallel to the yz-plane. Draw this path in your picture.

Solution. The black curve in Figure 3 is the path.

(d) Find a parametrization $\mathbf{r}(t)$ of the ant's path along the portion of the sphere shown in your picture. Specify the domain for $\mathbf{r}$, i.e. the initial time when the ant is at P and the final time when it hits the xy-plane.

Solution. $x = 1$ along the path and $f(1, y) = \sqrt{4 - y^2}$, so setting $y = t$ we get the parametrization

$$\mathbf{r}(t) = (1, t, \sqrt{4 - t^2}).$$

where $1 \le t \le 2$ so that $\mathbf{r}(1) = P$ and $\mathbf{r}(2) = (1, 2, 0)$ is when the ant reaches the xy-plane.

4. Suppose C is a plane curve parameterized by $\mathbf{r}\colon [0, 4] \to \mathbb{R}^2$. Consider $\mathbf{f}\colon [0, 2] \to \mathbb{R}^2$ defined by $\mathbf{f}(s) = \mathbf{r}(s^2)$.

(a) Explain why $\mathbf{f}$ also parameterizes C. Why is the domain of $\mathbf{f}$ not $[0, 4]$?

Solution. For any s in $[0, 2]$, one has s^2 is in $[0, 4]$ and so $\mathbf{f}(s) = \mathbf{r}(s^2)$ is indeed a point on C. Moreover, as s varies over $[0, 2]$, the value s^2 varies over all of $[0, 4]$, so we get the whole of C as the image of $\mathbf{f}$. The domain of $\mathbf{f}$ is not $[0, 4]$ because any $s \in (2, 4]$ has $s^2 > 4$ and so s^2 is not in the domain of $\mathbf{r}$.

(b) Calculate $\left|\mathbf{f}'(s)\right|$ in terms of $\left|\mathbf{r}'(t)\right|$. Check your answer with the instructor.

Solution. Suppose $\mathbf{r}(t) = \left(r_1(t),\ r_2(t)\right)$ where $r_i\colon\ [0,4] \to \mathbb{R}$ so that

$$\mathbf{f}(s) = \mathbf{r}(s^2) = \left(r_1(s^2),\ r_2(s^2)\right)$$

By the 1-variable chain rule, the derivative of $\mathbf{f}$ is then

$$\mathbf{f}'(s) = \left(r_1'(s^2)\cdot 2s,\ r_2'(s^2)\cdot 2s\right).$$

Hence the velocity at time s is

$$\left|\mathbf{f}'(s)\right| = \sqrt{\left(r_1'(s^2)\cdot 2s\right)^2 + \left(r_s'(s^2)\cdot 2s\right)^2} = \sqrt{\left(\left(r_1'(s^2)\right)^2 + \left(r_s'\left(s^2\right)\right)^2\right)(2s)^2} = \left|\mathbf{r}'(s^2)\right|\cdot 2s$$

since $s \geq 0$ implies $\sqrt{(2s)^2} = 2s$.

(c) Use your answer in (b) to confirm that using either $\mathbf{r}$ or $\mathbf{f}$ gives the same answer for the length of C.

Solution. In terms of $\mathbf{f}$ the length is

$$\int_0^2 \left|\mathbf{f}'(s)\right|\ ds = \int_0^2 \left|\mathbf{r}'(s^2)\right|\cdot(2s)\ ds = \int_0^4 \left|\mathbf{r}'(u)\right|\ du$$

where we did the substitution $u = s^2$ in the last step. Now replacing u by t in $\int_0^4 \left|\mathbf{r}'(u)\right|\ du$ gives the formula for the length of C in terms of $\mathbf{r}$, as desired.